Lecture Notes on Syzygies

From Linear Dependence to Relations in Commutative Algebra

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1 Motivation from linear algebra

In linear algebra, linear dependence of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_r \in V$ is expressed by an equation

$$c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r = \mathbf{0}$$

for some scalars $c_1, \ldots, c_r \in \mathbb{F}$.

In commutative algebra, one studies an analogous notion for elements of a ring, but now the coefficients are allowed to lie in the ring R itself.

This is the basic idea behind the notion of a *syzygy*.

Guiding question

Suppose $f_1, \ldots, f_r \in R$. Can we find elements $a_1, \ldots, a_r \in R$ such that

$$a_1f_1 + \cdots + a_rf_r = 0?$$

If yes, then the tuple $(a_1, \ldots, a_r)$ records a relation among the f_i. That relation is called a syzygy.

2 From linear dependence to syzygies

Let us compare the two situations.

2.1 Linear algebra

If V is a vector space over $\mathbb{F}$, then a relation among vectors

$$\mathbf{v}_1, \ldots, \mathbf{v}_r \in V$$

has the form

$$c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r = \mathbf{0}, \qquad c_i \in \mathbb{F}.$$

The coefficients come from the field $\mathbb{F}$.

2.2 Commutative algebra

If R is a commutative ring and

$$f_1, \ldots, f_r \in R,$$

then a relation among these elements has the form

$$a_1 f_1 + \cdots + a_r f_r = 0, \qquad a_i \in R.$$

The coefficients now come from the ring R itself.

Remark 2.1. This is richer than linear algebra because coefficients are no longer just constants; they may be polynomials or other ring elements.

3 Definition of syzygy

Definition 3.1. Let R be a commutative ring, and let $f_1, \ldots, f_r \in R$. A *syzygy* among $f_1, \ldots, f_r$ is a tuple

$$(a_1, \ldots, a_r) \in R^r$$

such that

$$a_1 f_1 + \cdots + a_r f_r = 0.$$

Definition 3.2. The set of all syzygies among $f_1, \ldots, f_r$ is called the *syzygy module* of $f_1, \ldots, f_r$, denoted

$$\operatorname{Syz}(f_1, \ldots, f_r) = \{(a_1, \ldots, a_r) \in R^r \mid a_1 f_1 + \cdots + a_r f_r = 0\}.$$

Proposition 3.3. $\operatorname{Syz}(f_1, \ldots, f_r)$ *is an* R*-submodule of* R^r*.*

Proof. Let $(a_1, \ldots, a_r)$ and $(b_1, \ldots, b_r)$ be syzygies. Then

$$a_1 f_1 + \cdots + a_r f_r = 0, \qquad b_1 f_1 + \cdots + b_r f_r = 0.$$

Adding these equations gives

$$(a_1 + b_1) f_1 + \cdots + (a_r + b_r) f_r = 0,$$

so the sum is again a syzygy.

If $c \in R$, then multiplying the first equation by c gives

$$(c a_1) f_1 + \cdots + (c a_r) f_r = 0.$$

So scalar multiples are again syzygies. □

4 The first basic example

Let

$$R = \mathbb{C}[x, y].$$

Choose

$$f_1 = x, \qquad f_2 = y.$$

A syzygy of f_1, f_2 is a pair $(a_1, a_2) \in R^2$ such that

$$a_1 x + a_2 y = 0.$$

Example 4.1. The pair

$$(-y, x)$$

is a syzygy of x and y, because

$$(-y)x + xy = -xy + xy = 0.$$

This is the first example one should remember.

Interpretation

The pair $(-y, x)$ means that the generators x and y are not independent. There is a precise way to combine them so that they cancel:

$$(-y) \cdot x + x \cdot y = 0.$$

5 Why a syzygy is a tuple in R^r

Suppose we have chosen elements

$$f_1, \dots, f_r \in R.$$

A relation among them must specify one coefficient for each f_i. Therefore the data of a relation is naturally a tuple

$$(a_1, \dots, a_r) \in R^r.$$

The equation

$$a_1 f_1 + \cdots + a_r f_r = 0$$

says that this tuple produces the zero relation.

So:

- R^r is the space of all coefficient choices,
- a syzygy is a coefficient choice that gives zero.

6 Syzygies as kernels

This is the cleanest conceptual viewpoint.

Given $f_1, \dots, f_r \in R$, define an R-linear map

$$\varphi : R^r \to R$$

by

$$\varphi(a_1, \dots, a_r) = a_1 f_1 + \cdots + a_r f_r.$$

Then by definition,

$$\mathrm{Syz}(f_1, \dots, f_r) = \ker(\varphi).$$

Thus a syzygy is simply an element of the kernel of the map determined by the chosen generators.

Remark 6.1. This is the most penetrating viewpoint:

$$\text{syzygy} = \text{kernel element of the generator map.}$$

6.1 The example x, y

Let

$$\varphi : \mathbb{C}[x, y]^2 \to \mathbb{C}[x, y], \qquad \varphi(a_1, a_2) = a_1 x + a_2 y.$$

Then

$$\mathrm{Syz}(x, y) = \ker(\varphi).$$

Since

$$\varphi(-y, x) = (-y)x + xy = 0,$$

the pair $(-y, x)$ lies in the kernel.

7 Relation with ideals

If $f_1, \dots, f_r \in R$, they generate the ideal

$$(f_1, \dots, f_r) = \{a_1 f_1 + \cdots + a_r f_r \mid a_i \in R\}.$$

So there is a natural surjective map

$$\pi : R^r \to (f_1, \dots, f_r), \qquad (a_1, \dots, a_r) \mapsto a_1 f_1 + \cdots + a_r f_r.$$

The kernel of this map is the module of syzygies:

$$0 \to \mathrm{Syz}(f_1, \dots, f_r) \to R^r \xrightarrow{\pi} (f_1, \dots, f_r) \to 0.$$

This exact sequence says:

- R^r remembers the chosen generators,
- the image is the ideal they generate,
- the kernel records the hidden relations among them.

8 All syzygies of x and y

We now study the example

$$R = k[x, y], \qquad f_1 = x, \qquad f_2 = y$$

more carefully.

Suppose

$$ax + by = 0$$

for some $a, b \in R$. We claim that every such relation is a multiple of $(-y, x)$.

Proposition 8.1. *In $R = k[x, y]$,*

$$\mathrm{Syz}(x, y) = R \cdot (-y, x).$$

That is, every syzygy of x and y has the form

$$(-cy, cx)$$

for some $c \in R$.

Proof. Suppose
$$ax + by = 0.$$
Then
$$ax = -by.$$
Hence ax is divisible by y. Since x and y are relatively prime in $k[x, y]$, it follows that a must be divisible by y. So we may write
$$a = -cy$$
for some $c \in R$. Substituting gives
$$(-cy)x + by = 0,$$
hence
$$y(-cx + b) = 0.$$
Since R is an integral domain,
$$-cx + b = 0,$$
so
$$b = cx.$$
Therefore
$$(a, b) = (-cy, cx) = c(-y, x).$$
□

Remark 8.2. This is a beautiful example: there are infinitely many syzygies, but they are all generated by one basic syzygy.

9 A second example

Let
$$R = k[x, y]$$
and consider
$$f_1 = x^2, \qquad f_2 = xy, \qquad f_3 = y^2.$$
We look for triples $(a, b, c) \in R^3$ such that
$$ax^2 + bxy + cy^2 = 0.$$

Two obvious syzygies are
$$(-y, x, 0)$$
and
$$(0, -y, x).$$

Indeed,
$$(-y)x^2 + x(xy) + 0 \cdot y^2 = -x^2y + x^2y = 0,$$
and
$$0 \cdot x^2 + (-y)(xy) + x(y^2) = -xy^2 + xy^2 = 0.$$
Remark 9.1. This shows how syzygies arise from different ways of forming the same product.

10 Syzygies and presentations

Suppose M is an R-module generated by elements

$$m_1, \dots, m_r.$$

Then there is a surjective map

$$\pi : R^r \to M, \qquad (a_1, \dots, a_r) \mapsto a_1 m_1 + \cdots + a_r m_r.$$

Its kernel is the module of relations among the generators.

Definition 10.1. The kernel of the map $R^r \to M$ is called the *module of first syzygies* of the chosen generators of M.

Thus one has an exact sequence

$$0 \to K_1 \to R^r \to M \to 0,$$

where

$$K_1 = \ker(R^r \to M).$$

This is called a *presentation* of M: the module M is described by generators and relations.

11 Higher syzygies

The first syzygy module may itself have generators and relations.

Suppose

$$0 \to K_1 \to F_0 \to M \to 0$$

with F_0 free. If K_1 is generated by some elements, then there is a surjection

$$F_1 \to K_1 \to 0$$

from another free module F_1. The kernel of $F_1 \to K_1$ consists of relations among the first syzygies.

These are called *second syzygies*.

Continuing in this way, one gets

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0.$$

Definition 11.1. An exact sequence

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0$$

with each F_i a free R-module is called a *free resolution* of M.

Interpretation:

Module	Meaning
F_0	generators of M
F_1	first syzygies
F_2	second syzygies
F_3	third syzygies
$\vdots$	$\vdots$

12 A concrete exact sequence

For the ideal

$$I = (x, y) \subseteq k[x, y],$$

consider the map

$$\varphi : R^2 \to I, \qquad \varphi(a, b) = ax + by.$$

This map is surjective, and its kernel is generated by $(-y, x)$. Therefore we have an exact sequence

$$0 \to R \xrightarrow{\psi} R^2 \xrightarrow{\varphi} I \to 0,$$

where

$$\psi(c) = (-cy, cx).$$

This says:

- the ideal I has two generators x, y,
- these generators satisfy one basic relation,
- all other relations come from this one.

13 Comparison with linear algebra

It is worth returning to the analogy with linear algebra.

13.1 Vector spaces

A map

$$T : \mathbb{F}^r \to V, \qquad (c_1, \dots, c_r) \mapsto c_1\mathbf{v}_1 + \cdots + c_r\mathbf{v}_r$$

has kernel equal to the space of linear relations among $\mathbf{v}_1, \dots, \mathbf{v}_r$.

13.2 Modules over rings

A map

$$\varphi : R^r \to M, \qquad (a_1, \dots, a_r) \mapsto a_1m_1 + \cdots + a_rm_r$$

has kernel equal to the module of syzygies among $m_1, \dots, m_r$.

So syzygies are the natural generalization of linear dependence from vector spaces to modules.

14 Why syzygies matter

Syzygies matter because they reveal hidden structure.

Knowing generators of an ideal or module is only the first step. One also wants to know:

- how the generators depend on one another,

- how complicated those dependencies are,
- and whether there are higher layers of dependencies.

Thus syzygies are a way of measuring structure beyond generators.

Slogan

$$\text{generators} \longrightarrow \text{relations} \longrightarrow \text{relations among relations} \longrightarrow \cdots$$

15 A brief geometric remark

In algebraic geometry, one often studies a variety by the equations defining it inside projective space. Those equations may satisfy relations among themselves. These relations are syzygies.

So:

- equations describe the variety,
- syzygies describe how the equations fit together.

This is why syzygies are central in modern algebraic geometry.

16 Exercises

Exercise 16.1. Let $R = \mathbb{C}[x, y]$. Show directly that

$$(-y, x)$$

is a syzygy of x and y.

Exercise 16.2. Let $R = k[x, y]$. Prove that every syzygy of x and y is a multiple of $(-y, x)$.

Exercise 16.3. Find two explicit syzygies among

$$x^2,\ xy,\ y^2.$$

Exercise 16.4. Let $R = k[x, y, z]$. Show that

$$(-y, x, 0), \qquad (-z, 0, x), \qquad (0, -z, y)$$

are syzygies of x, y, z.

Exercise 16.5. Let $M = R/(x, y)$ with $R = k[x, y]$. Write down a presentation of M and describe its first syzygy module.

Exercise 16.6. Explain why $\mathrm{Syz}(f_1, \ldots, f_r)$ is naturally an R-submodule of R^r.

17 Solutions sketch

Exercise 1

Compute

$$(-y)x + xy = -xy + xy = 0.$$

So $(-y, x)$ is a syzygy.

Exercise 2

If

$$ax + by = 0,$$

then $ax = -by$. Since x and y are relatively prime, $y \mid a$. Write $a = -cy$. Substituting gives $b = cx$. Hence

$$(a, b) = c(-y, x).$$

Exercise 3

Two syzygies are

$$(-y, x, 0), \qquad (0, -y, x).$$

Exercise 4

Check directly that each tuple gives zero when substituted into

$$ax + by + cz.$$

Exercise 5

The quotient $M = R/(x, y)$ has presentation

$$R^2 \to R \to R/(x, y) \to 0,$$

where $R^2 \to R$ sends the standard basis vectors to x and y. The first syzygy module is generated by $(-y, x)$.

Exercise 6

One checks closure under addition and scalar multiplication exactly as in the proof above.

18 Final summary

A syzygy is a relation among chosen elements of a ring or module.

The transition from linear algebra to commutative algebra is simple but profound:

- in linear algebra, coefficients come from a field,

- in commutative algebra, coefficients come from the ring itself.

Thus, if $f_1, \ldots, f_r \in R$, a syzygy is a tuple

$$(a_1, \ldots, a_r) \in R^r$$

such that

$$a_1 f_1 + \cdots + a_r f_r = 0.$$

The example

$$(-y)x + xy = 0$$

already contains the whole philosophy: generators are not enough by themselves; one must also understand the relations tying them together.

That is what syzygies measure.